Validate Bst

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The problem

- Every node must be strictly greater than all left-subtree values and <= all right-subtree values.

Intuition

- Every node must lie within bounds inherited from its ancestors, not just compared with its parent.

Approach

- 1. Start with minus and plus infinity
- 2. Check the node is within bounds
- 3. Left child tightens the upper bound
- 4. Right child tightens the lower bound

Java solution

```java
public class ValidateBst {
    public static boolean validateBst(BST t) { return valid(t, Long.MIN_VALUE, Long.MAX_VALUE); }

    private static boolean valid(BST t, long min, long max) {
        if (t == null) return true;
        if (t.value < min || t.value >= max) return false;
        return valid(t.left, min, t.value) && valid(t.right, t.value, max);
    }
}
```

Complexity

- Time: O(n)
- Space: O(h)

Recap

- Every node must lie within bounds inherited from its ancestors, not just compared with its parent.
- Complexity: O(n)